

White Paper: Proof of Social Learning Protocol

Abstract

This white paper introduces a novel **Proof of Social Learning Protocol (PSLP)** designed to transparently and immutably record knowledge transfer and learning events within communities of practice using Distributed Ledger Technology (DLT), specifically a blockchain. Unlike traditional blockchain protocols that rely on computational proof-of-work for validation, PSLP leverages **social learning events as the fundamental validation mechanism** ("mining"). The protocol is particularly well-suited for communities focused on procedural knowledge, such as the Liberating Structures community of practice, where specific methods or "structures" can be seen as **discreet packets of procedural knowledge**. A core component is a **shared Learning Journal (wiki)** interface that uses recorded learning progress, community evaluation, and available data to enhance user learning and facilitate minimally resource-intensive information organization. **Privacy, data sovereignty, security, ethical sharing, attribution, and the cultivation of a regenerative economy of value, discovery, divestment, and distribution** are paramount to the protocol's design.

1. Introduction: The Value of Capturing Social Learning

Knowledge transfer and learning within communities are often implicit, fragmented, and difficult to track or formally recognize. While digital platforms exist for sharing information, they frequently rely on centralized models that can lead to silos, data exploitation, and a lack of trust. Traditional credentialing systems can be costly, exclusive, and may not accurately reflect embodied practice or collaborative learning.

Blockchain technology offers a decentralized and transparent means of recording transactions and data. However, many blockchain applications have focused primarily on financial value or speculative assets, often reproducing existing inequalities and power dynamics. There is a need to explore how DLT can be applied to support social and community-driven objectives, moving beyond purely transactional exchanges to enable social and solidarity practices.

Capturing and valuing social learning requires new approaches that align with the principles of decentralized collaboration, community empowerment, and the recognition of diverse forms of value beyond the purely monetary. This white paper proposes a Proof of Social Learning Protocol (PSLP) as a framework for building such a system.

2. The Liberating Structures Community: A Suitable Use Case

The Liberating Structures (LS) community of practice provides an ideal context for piloting a Proof of Social Learning Protocol. LS are a set of forty-three microstructures that can be used to engage groups of any size in collaborative problem-solving, innovation, and planning. They are

simple rules to unleash a culture of innovation and make it practical for people at any skill or hierarchical level to quickly become expert contributors.

Importantly, each Liberating Structure is a **discreet packet of procedural knowledge**, defined by five design elements: how participation is distributed, how groups are formed, the steps over time, the invitation (question/prompt), and how materials and space are organized. Learning LS involves direct experience, practice, and debriefing with peers. The community emphasizes shared learning, mutual support, and co-creation.

The LS framework and its community embody key principles that align with PSLP's goals:

- **Focus on Process:** LS emphasizes how people interact and collaborate, providing a structured approach to social dynamics.
- **Community of Practice:** Learning and spreading LS occurs through shared experiences and peer-to-peer interaction.
- **Open Source Nature:** LS is licensed under Creative Commons, encouraging adaptation and sharing.
- **Diverse Participation:** LS methods are designed to include and unleash everyone, regardless of hierarchy or skill level.

Documenting the application and learning of these structured interactions provides a tangible starting point for recording complex social learning events on a ledger.

3. The Proof of Social Learning Protocol (PSLP)

3.1 Core Concept: A Decentralized Ledger for Learning

The PSLP utilizes a DLT, specifically a blockchain, as an open, shared, decentralized database to record information about social learning events. This ledger provides a transparent and immutable record of learning activities within the community.

3.2 Social Learning Events as "Mining"

Instead of relying on energy-intensive computational puzzles for validation, as in Proof-of-Work, **PSLP uses social learning events themselves as the "mining" process** that validates and adds data to the ledger.

A social learning event could be:

- A practice session where a specific LS is used.
- A workshop or training where LS methods are taught and practiced.
- A coaching session applying LS-based inquiry.
- Any structured interaction where explicit learning or knowledge transfer using defined protocols occurs.

During or after a social learning event, participants collectively, or designated stewards/facilitators, submit a record of the event to the network. The validation of this record is based on the **social consensus** of the participants involved, potentially combined with community-based attestation or evaluation processes. For example, participants could cryptographically sign off on the record of the event, or a quorum of participants could attest to its occurrence and basic details. This distributed trust mechanism validates the "block" of learning data before it is added to the chain. This aligns with the idea of **distributing trust** rather than removing it entirely.

3.3 Data Model: Recording Learning Activities

Each social learning event recorded on the PSLP ledger would include structured data, potentially captured through intuitive interfaces to minimize user effort and resource intensity. The data model would encompass:

- **Context:** Where and when the learning event took place (e.g., organization, community group, date, time).
- **Objective:** The stated purpose or goal of the learning event. For LS applications, this could link to the "Nine Whys" of the work.
- **Participants:** Pseudonymized or verified identities of individuals who participated. Permission layers can manage the visibility of this data based on user preferences and community norms.
- **Process/Structure:** Details about the specific LS or sequence of structures ("strings") used, including time allocation and group configurations.
- **Content/Outcomes:** Summaries of discussions, key insights, emergent ideas, decisions made, or planned next steps. This could include links to external documentation or artifacts where appropriate.
- **Reflection:** Evaluation of Return on Time Invested, structure choice, design, novelty/innovation, second thoughts, Tips and Traps.

Capturing this structured data allows for the mapping of how different structures are applied in various contexts for specific purposes, and the outcomes they generate.

3.4 The Shared Learning Journal (Wiki Interface)

The shared Learning Journal serves as the primary interface for users to access and interact with the data on the PSLP ledger. Designed as a collaborative wiki, it visualizes the recorded learning events and offers functionalities to support user learning and community knowledge building.

The journal leverages the structured data on the ledger in several ways:

- **Learning Progress:** Users can track their own participation in learning events and visualize their experience with different LS patterns, potentially using frameworks like the Ecocycle or developmental pathways. This progress can influence access permissions within the journal and Reflections of Social Learning Events

- **Novelty and Utility:** The journal can highlight instances where structures were used in novel contexts or situations, or where community evaluation indicates high utility or impact.
- **Organization and Navigation:** The wiki structure allows for organizing information about LS patterns, strings, tips, traps, riffs, variations, and examples based on the recorded data. It can use metadata (context, objective, participants) to facilitate searching and filtering, enabling users to find relevant examples or connect with practitioners.
- **Minimally Resource Intensive:** By drawing structured data from the ledger, the wiki can automate certain organizational tasks, reducing the manual effort required for content management compared to traditional wikis. Non-intrusive interfaces for data input during learning events could also minimize participant burden.

The journal fosters a **knowledge commons** where collective wisdom is shared and made accessible. It can be enhanced by community evaluation features, allowing users to rate or comment on the documented events or the insights derived from them, further supporting the organization and highlighting of valuable information.

3.5 Permissions and Access Control

PSLP incorporates a granular permission system, managed on the ledger and reflected in the Learning Journal, to ensure **data sovereignty** and control over who can access and share information. Permissions are dynamically managed based on factors such as:

- **User Identity:** Individuals control their own data, deciding which information is publicly shared, shared within specific community circles, or kept private.
- **Learning Progress/Role:** As users gain experience or take on specific roles within the community (e.g., facilitator, steward), they may be granted permissions to access or contribute to different levels of information in the journal, such as design details or aggregated data.
- **Community Evaluation/Trust:** Reputation or trust built through participation and positive community evaluation can unlock higher levels of access or influence over content organization.

This ensures that while the core record of learning events is immutable on the ledger, the accessibility and visibility of detailed information within the user-facing journal are governed by community norms, trust, and individual consent.

3.6 Ethical Sharing, Attribution, and the Economy of Value

The PSLP is designed to promote **ethical sharing and proper attribution** of contributions. By recording the participants and process of learning events, the protocol provides a basis for crediting individuals and groups for their roles in knowledge creation and transfer.

The protocol aims to cultivate a regenerative **economy of value, discovery, divestment, and distribution** within the learning community. This goes beyond traditional monetary exchange, recognizing and valuing diverse contributions such as:

- **Value Creation:** Documenting the successful application of structures or the emergence of new insights acknowledges the value created through collaboration.
- **Discovery:** The journal facilitates the discovery of effective practices, novel applications, and potential collaborators.
- **Distribution:** Insights, knowledge, and potentially related resources or opportunities can be distributed throughout the network.
- **Divestment:** While not focused on financial divestment, the system could, for example, be used to distribute financial resources throughout the community of practice.

By making contributions visible and building a system of reputation and attribution based on social consensus and community evaluation, the protocol incentivizes participation and sharing. This supports a **commons-based collaborative economy** centered on shared resources and reciprocal exchange.

3.7 Privacy, Data Sovereignty, and Security

Ensuring **privacy, data sovereignty, and security** is critical. The inherent transparency of blockchain must be balanced with the need for confidentiality and user control.

- **Privacy:** While transaction details (records of learning events) are on-chain, sensitive data can be managed off-chain with references on the ledger, or encrypted using techniques like Zero-Knowledge Proofs (ZKPs) or Trusted Execution Environments (TEEs) where feasible. Pseudonymity or anonymization of participant data can also be employed.
- **Data Sovereignty:** Users maintain control over their personal learning data and identity, deciding how it is shared and used in accordance with their preferences and community agreements. This aligns with the principle of "nothing about us without us".
- **Security:** The DLT provides inherent security through cryptographic proof and distributed consensus, protecting the integrity and immutability of the learning records. Robust access controls based on cryptography can protect sensitive information.

Developing this aspect requires careful design, potentially leveraging advancements in privacy-preserving technologies within the DLT space.

4. Challenges and Future Work

Implementing a Proof of Social Learning Protocol presents several challenges:

- **Technological Complexity:** Designing and building the DLT and the wiki interface with the required privacy, security, and permission features is complex. User-friendly interfaces are needed to make the technology accessible to a broad community.

- **Social Adoption and Cultural Shift:** Implementing such a system requires a cultural shift towards more explicit documentation, sharing, and collaborative validation of learning. Acknowledging resistance or apprehension from new users is essential.
- **Defining and Measuring Learning:** While LS offers discreet structures, capturing the nuances of embodied practice, experiential learning, and emergent outcomes remains challenging. Developing community-agreed metrics and evaluation processes is key.
- **Sustainability and Governance:** Establishing a sustainable model for the development and maintenance of the protocol and the Learning Journal, including decentralized governance mechanisms, is crucial.
- **Legal and Regulatory Landscape:** Navigating the evolving legal status of decentralized organizations and the use of DLT for non-financial purposes is necessary.

Future work will involve refining the protocol design, building and testing a minimum viable product, and engaging the LS community in a participatory design process. Research into integrating advanced privacy features, exploring different DLT platforms, and developing user-friendly interfaces will be essential.

5. Conclusion

The Proof of Social Learning Protocol offers a compelling vision for leveraging DLT to support and amplify social learning within communities of practice like Liberating Structures. By using social learning events as the validation mechanism, the protocol aligns the technology with the community's core activities and values. The integrated Learning Journal can become a powerful tool for knowledge sharing, skill development, and fostering a regenerative economy of value based on collaboration, attribution, and the ethical distribution of insights. While significant technical and social challenges remain, the potential for building a more transparent, equitable, and vibrant learning ecosystem makes this a worthwhile endeavor.